

PROPOSAL FOR FEDERAL GOVERNMENT DIGITALIZATION MANDATE

Digital Procurement & Contractor Management Portal

In alignment with the Federal Government's directive to digitalize all parastatal operations

Sea View Properties Ltd · A Subsidiary of the Nigerian Ports Authority

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SVP

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1. Executive Summary

In alignment with the Federal Government of Nigeria's directive to digitalize operations across all parastatals and government agencies, we propose the implementation of a comprehensive **Digital Procurement & Contractor Management Portal** for Sea View Properties, a subsidiary of the Nigerian Ports Authority.

This portal replaces manual, paper-based procurement processes with a secure, transparent, and auditable digital system. It covers the entire procurement lifecycle — from contractor registration to contract award, project completion, audit review, and payment processing — while enforcing multi-tier approval workflows, role-based access control, and immutable audit logging.

The system is built on modern, enterprise-grade technology and is designed to **eliminate corruption, reduce processing times, ensure accountability, and provide real-time visibility** into all procurement activities.

2. Background & Problem Statement

2.1 Current Challenges

The existing procurement process at Sea View Properties and similar parastatals suffers from:

- **Manual, paper-based workflows** that are slow, error-prone, and difficult to track
- **Lack of transparency** in contract awards, creating opportunities for corruption and favouritism
- **No centralized audit trail**, making it difficult to investigate disputes or misconduct
- **Fragmented communication** between contractors, procurement officers, and approval authorities
- **No real-time reporting**, preventing leadership from making data-driven decisions
- **Difficulty in tracking contractor performance** and project completion status
- **Inefficient payment processes** with no digital evidence or accountability

2.2 Government Mandate

The Federal Government of Nigeria has mandated all parastatals to digitalize their operations to:

- Improve service delivery and operational efficiency
- Enhance transparency and accountability in public procurement
- Reduce corruption and financial leakages
- Enable data-driven governance and reporting
- Align with international best practices in public sector management

3. Proposed Solution

3.1 Overview

The **Digital Procurement & Contractor Management Portal** is a web-based platform that digitizes the end-to-end procurement and contractor management lifecycle. It provides role-based access to all stakeholders and enforces a structured, multi-tier approval workflow that mirrors existing organizational hierarchies.

3.2 Technology Stack

Layer	Technology
Frontend	Next.js 15, TypeScript, Tailwind CSS, shadcn/ui
Backend / Database	Supabase (PostgreSQL, Authentication, Storage)
Security	Supabase Row Level Security (RLS), Role-based Access Control
PDF Generation	@react-pdf/renderer for award letters and official documents
Real-time Notifications	Supabase Realtime, Sonner in-app notifications
Hosting	Cloud-based (Vercel / Supabase Cloud)

3.3 Key Modules

Contractor Registration

Online self-service registration for contractors. Document upload (CAC certificate, tax clearance, specialized competence certificates). Verification and approval by the MD. Digital contractor profile management.

Proposal Submission

Contractors submit proposals online with supporting documents. Automatic tracking and unique reference numbers. Status visibility for contractors at every stage.

Multi-Tier Approval Workflow

The system enforces a strict, sequential approval chain: Contractor submits → MD Initial Review → Procurement Officer Appraisal → Head of Procurement Approval → MD Final Approval → Contract Awarded. No stage can be skipped. Each approver can approve, reject, or request modifications.

Digital Signature & Contract Award

Authorized officers upload digital signatures (PNG). System auto-generates professional PDF award letters with embedded signatures. Unique contract reference numbers (CON-YYYY-XXXX). Award letters downloadable and archived.

Project Completion Management

Contractors submit completion reports with evidence. MD verification → Head of Audit review → Forwarding to Accounts for payment processing.

Audit & Compliance

Immutable audit log — database triggers prevent UPDATE/DELETE of log entries. Every action is logged with timestamp, user, and action details. ICT Administrator has read-only access. Internal minutes hidden from contractors.

Accounts & Payment Processing

Head of Accounts processes payments upon audit approval. Payment evidence upload (receipts, transfer confirmations). Unique payment reference numbers (PAY-YYYY-XXXX). Contractors track payment status in real-time.

Executive Dashboard

Real-time analytics and KPIs for leadership. Visual charts: total proposals, active contracts, completed projects, payments processed. Mobile-responsive design. Role-specific dashboard views.

Universal Search

Fuzzy search across all entities (contractors, proposals, contracts, payments). Quick access to any record from any page. Search by name, reference number, status, or date.

Notifications

Real-time in-app notifications via Supabase Realtime. Users alerted when actions are required. Notification history and read/unread status.

4. User Roles & Access Control

Access control is enforced at the **database level** using Supabase Row Level Security (RLS), ensuring that no user can access data outside their authorized scope — even through direct database queries.

Role	Access Level
Contractor	Register, submit proposals, upload completions, view payment status
Managing Director (MD)	Full access — initial & final approval, verification
Procurement Officer	Appraise proposals, forward to Head of Procurement
Head of Procurement	Review appraisals, approve/reject, forward to MD
Head of Audit	Review completions, approve, forward to Accounts
Head of Accounts	Process payments, upload payment evidence
ICT Administrator	User management, audit logs, system settings (cannot participate in approvals)

5. Security & Compliance

5.1 Data Security

- **Row Level Security (RLS)** policies on every table
- Role-based access control enforced at database and application layers
- Secure file storage with bucket-level policies
- Environment-protected service keys

5.2 Audit & Accountability

- **Immutable audit log** — tamper-proof by design (database triggers prevent modification)
- Complete action history for every record
- User activity tracking (login times, actions performed)
- ICT Administrator oversight without procurement interference

5.3 Data Integrity

- PostgreSQL relational database with foreign key constraints
- Enumerated types for statuses (prevents invalid states)
- Automatic timestamping on all records
- Unique reference number generation for contracts and payments

6. Benefits & Value Proposition

6.1 For the Federal Government

- **Transparency:** Every procurement action is logged, traceable, and auditable
- **Anti-Corruption:** Multi-tier approvals with no bypass capability; immutable audit trail
- **Efficiency:** Digital workflows reduce processing time from weeks to days
- **Data-Driven Governance:** Real-time dashboards for monitoring and reporting
- **Alignment with Digital Economy Agenda:** Supports the national digitalization mandate

6.2 For Sea View Properties / NPA

- **Cost Savings:** Eliminates paper, printing, and manual processing costs
- **Reduced Processing Time:** Parallel and tracked workflows
- **Improved Contractor Relationships:** Transparency and self-service portal
- **Risk Mitigation:** Complete audit trail for investigations and compliance
- **Scalability:** System can be extended to other NPA subsidiaries

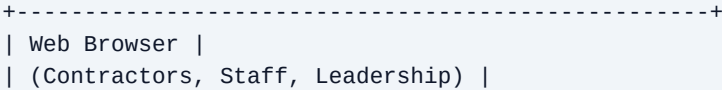
6.3 For Contractors

- **Convenience:** Online registration, proposal submission, and status tracking
- **Transparency:** Real-time visibility into proposal and payment status
- **Fairness:** Standardized process with no human interference in workflow

7. Project Implementation Plan

Phase	Timeline	Activities
Phase 1: Deployment & Configuration	Weeks 1–2	Deploy portal to production, configure Supabase, set up user accounts, migrate existing data
Phase 2: User Training & Onboarding	Weeks 2–3	Train all staff roles, train ICT Administrator, provide contractor onboarding guide
Phase 3: Go-Live & Support	Week 4	Official launch, migration from manual to digital, dedicated support
Phase 4: Enhancement & Scaling	Ongoing	Collect feedback, add features, scale to additional NPA subsidiaries

8. Technical Architecture



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| HTTPS
+-----v-----+
| Next.js 15 Application |
| (SSR + Client-side, Role-based Routing) |
| Tailwind CSS + shadcn/ui Components |
+-----+-----+
|
+-----v-----+
| Supabase Backend |
| +-----+ +-----+ +-----+ |
| |PostgreSQL| | Auth | | Storage | |
| | + RLS | | (JWT) | | (Documents) | |
| +-----+ +-----+ +-----+ |
| +-----+ |
| | Realtime (Notifications) | |
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```

9. System Requirements

9.1 User Requirements

- Modern web browser (Chrome, Firefox, Safari, Edge)
- Internet connection
- No software installation required (web-based)

9.2 Infrastructure Requirements

- Supabase account (Cloud or self-hosted)
- Vercel account for application hosting (or equivalent Node.js hosting)
- Domain name and SSL certificate
- Supabase environment variables (URL, Anon Key, Service Role Key)

10. Cost Considerations

Item	Description	Estimated Cost
Supabase	Database, Auth, Storage, Realtime	Free tier / Pro ~\$25/mo
Vercel Hosting	Application hosting	Free tier / Pro ~\$20/mo
Domain Name	Custom domain (e.g., procurement.seaview.npa.gov.ng)	~\$15/year
SSL Certificate	Included with Vercel/Supabase	Free
Development	One-time development cost	[To be quoted]
Training	User training and onboarding	[To be quoted]
Maintenance	Monthly support and updates (optional)	[To be quoted]

Note: The system uses open-source technologies, keeping licensing costs at zero. Ongoing costs are limited to cloud hosting and optional maintenance.

11. Why This Solution

11.1 Built for Nigerian Government Context

- Designed specifically for the NPA / Sea View Properties organizational structure
- Approval workflow mirrors existing hierarchical processes
- Role definitions match actual staff designations

11.2 Enterprise-Grade Yet Affordable

- Uses open-source technologies (zero licensing fees)
- Cloud-hosted (no server hardware required)
- Scales from dozens to thousands of users

11.3 Secure by Design

- Database-level security (RLS) — not just application-level
- Immutable audit log — tamper-proof
- Role separation (ICT Admin cannot interfere with procurement)

11.4 Rapidly Deployable

- System is already developed and tested
- Can be deployed within 2–4 weeks
- Minimal infrastructure setup required

12. Conclusion

The Digital Procurement & Contractor Management Portal represents a transformative step for Sea View Properties and the Nigerian Ports Authority in fulfilling the Federal Government's digitalization mandate. By replacing manual processes with a secure, transparent, and auditable digital system, the organization will achieve:

- **Full transparency** in all procurement activities
- **Accountability** through immutable audit trails
- **Efficiency** through automated workflows
- **Cost savings** through elimination of paper-based processes
- **Data-driven decision making** through real-time dashboards

We are confident that this solution will serve as a **model for digitalization across all NPA subsidiaries** and potentially other federal parastatals. We welcome the opportunity to present this proposal in detail and demonstrate the working system.

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This proposal is submitted in response to the Federal Government of Nigeria's directive on the digitalization of parastatal operations.